ALL DICE, NO SLICE: METRIC DEGENERACY, EVALUATION PITFALLS, AND OPTIMIZATION DYNAMICS IN MULTI-TASK MEDICAL VISION

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ABSTRACT

Joint classification and semantic segmentation via hard parameter-sharing networks offer an efficient paradigm for resource-constrained clinical AI deployments. While multi-task architectures frequently report high diagnostic accuracy and segmentation overlap, evaluating dual-task performance across diverse imaging modalities involves subtle metric aggregation effects, optimization trade-offs, and clinical task-formulation boundaries. This paper presents a systematic methodological investigation across four diverse clinical imaging modalities: brain tumor MRI, prostate histopathology, chest radiography, and multi-organ histology under patient-isolated cross-validation. We investigate four foundational research questions: first, we show analytically and empirically that macroscopic slice-averaged Dice scores establish an artificial performance baseline on negative-dominant cohorts that conceals complete foreground lesion omission; second, we examine representation boundaries on compressed biopsy images, clarifying how proxy morphological patch categorization diverges from whole-core clinical ordinal grading; third, we characterize multi-task optimization dynamics on shared encoders, demonstrating that dynamic gradient balancing can induce parameter instability and degrade classification discrimination; and fourth, we evaluate color deconvolution preprocessing, showing that standard optical density projection can attenuate discriminative cellular microtextures. We conclude with five practical evaluation guidelines to ensure reliable and clinically trustworthy translation of multi-task medical vision systems.

***Index Terms*—** Multi-Task Learning, Medical Image Analysis, Semantic Segmentation, Metric Evaluation, Digital Pathology.

1. INTRODUCTION

Deep learning systems in digital pathology and medical imaging are increasingly tasked with delivering both holistic diagnostic categorization and localized lesion delineation [1, 2]. While visual foundation models demonstrate high representation capacity across gigapixel whole-slide images [3–5], their computational footprint precludes direct deployment in edge environments or low-resource clinical facilities. Consequently, hard parameter sharing via lightweight encoder–decoder architectures, particularly dual-headed U-Net topologies [6], remains a compelling design pattern that couples classification and segmentation within a unified inference pass.

Dual-headed architectures employing lightweight backbones (such as VGG16 [7] and MobileNetV2 [8]) have been reported to achieve strong simultaneous classification and segmentation performance across diverse imaging modalities, including brain MRI, prostate biopsy, and chest radiography [9]. Because lightweight parameter-sharing networks offer substantial efficiency advantages over multi-network ensembles, understanding the empirical factors governing their performance is critical for clinical translation.

However, evaluating multi-task medical models involves complex interplay between metric definitions, task formulations, and optimization dynamics. First, conventional region-overlap metrics, such as macroscopic Dice scores, exhibit mathematical boundary artifacts in cohorts where lesion-free images predominate [10, 11]. Second, in computational pathology, the spatial resolution of inputs determines diagnostic feasibility; evaluating low-capacity encoders on heavily downsampled images may conflate coarse morphological tissue categorization with fine-grained clinical ordinal grading [12, 13]. Third, optimizing dense pixel-wise segmentation alongside global classification objectives frequently induces gradient interference on shared representations, demanding careful loss balancing [14, 15].

In this work, we conduct a systematic empirical inquiry into lightweight multi-task medical vision, structured around four core research questions:

1. **RQ1 (Metric Behavior in Sparse Regimes):** How do conventional overlap metrics behave when negative cases predominate, and how can true foreground lesion delineation be isolated from background-credit artifacts?
2. **RQ2 (Task Formulation vs. Resolution Boundaries):** What are the representation limits of standard 2D encoders on downsampled biopsy tiles, and how does performance differ between coarse morphological tissue proxies and whole-core clinical ordinal grading?
3. **RQ3 (Multi-Task Optimization Dynamics):** How do static loss weighting, dynamic gradient balancing (GradNorm), and lateral skip connections influence cross-task gradient competition on shared backbones?
4. **RQ4 (Preprocessing and Feature Preservation):** Does standard color deconvolution (Macenko stain normalization) preserve fine-grained diagnostic textures, or can optical density projection alter discriminative features?

To address these questions, we perform a controlled factorial benchmark evaluated under patient-isolated cross-validation across four diverse clinical datasets: TCGA-LGG brain MRI, PANDA prostate biopsies, SIIM-ACR pneumothorax radiographs, and PanNuke multi-organ histopathology (Figure 1). Our primary contributions are summarized as follows:

- We analyze the empty-mask Dice aggregation behavior analytically and empirically, demonstrating how macroscopic cohort averages establish a baseline floor on negative-dominant datasets and proposing disaggregated foreground reporting.
- We clarify the methodological distinction between proxy morphological tile classification and clinical ordinal grading in

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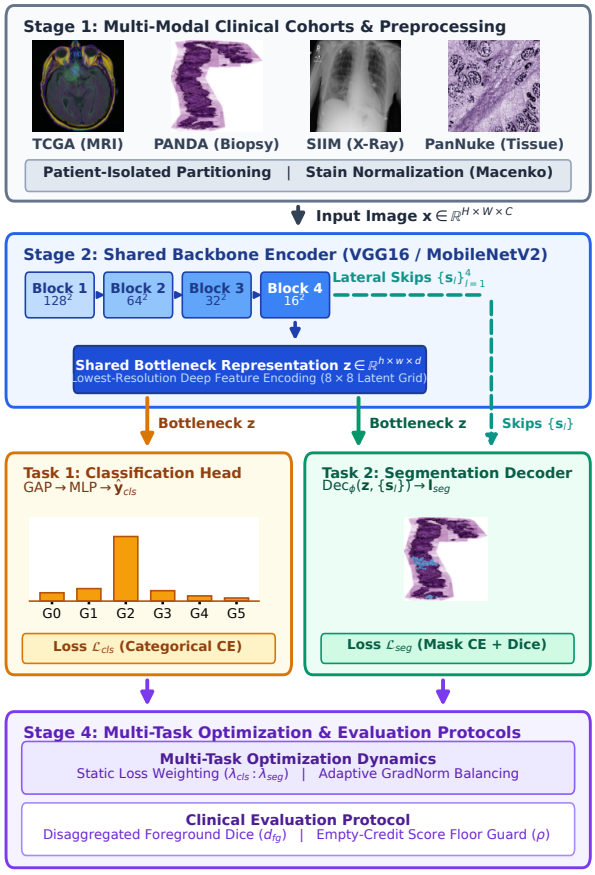


Fig. 1. System architecture and multi-task evaluation framework across four clinical imaging modalities, integrating shared feature extraction, decoupled diagnostic heads with lateral skip connections, and adaptive optimization dynamics under patient-isolated cross-validation.

digital pathology, delineating the spatial resolution boundaries of compact 2D networks.

- We systematically characterize multi-task optimization dynamics and preprocessing pipelines through controlled factorial ablations, quantifying the stability of static versus dynamic gradient balancing and evaluating the discriminative impact of stain normalization.

2. SYSTEM ARCHITECTURE AND METHODOLOGY

2.1. Architectural Formulation

Let $\mathbf{x} \in \mathbb{R}^{H\times W\times C}$ denote an input image. A shared convolutional backbone $f_\theta(\mathbf{x})$ maps the input to a bottleneck representation $\mathbf{z} \in \mathbb{R}^{h\times w\times d}$. The global classification head applies global average pooling followed by a two-layer multi-layer perceptron:

$$\hat{\mathbf{y}}_{cls} = \text{Softmax}\left(\mathbf{W}_2\,\phi\left(\mathbf{W}_1\,\text{GAP}(\mathbf{z}) + \mathbf{b}_1\right) + \mathbf{b}_2\right), \quad (1)$$

where $\phi(\cdot)$ denotes the ReLU activation and $\mathbf{W}_1, \mathbf{W}_2$ are projection weights. The segmentation branch passes $\mathbf{z}$ through a convolutional decoder Dec_ϕ augmented with lateral skip connections $\{\mathbf{s}_l\}_{l=1}^{L}$ from encoder stages, generating spatial logits $\mathbf{l}_{seg} \in \mathbb{R}^{H\times W\times M}$.

The composite multi-task training loss is defined as:

$$\mathcal{L}_{total} = \lambda_{cls}\mathcal{L}_{cls}(\mathbf{y}_{cls}, \hat{\mathbf{y}}_{cls}) + \lambda_{seg}\mathcal{L}_{seg}(\mathbf{y}_{seg}, \mathbf{l}_{seg}), \quad (2)$$

where $\mathcal{L}_{cls}$ is categorical cross-entropy with inverse-frequency class weights $w_c = N/(K\cdot N_c)$, and $\mathcal{L}_{seg}$ is binary or multi-class cross-entropy. In standard implementations, the Dice similarity coefficient is computed strictly post-hoc as an evaluation metric rather than an active loss component.

2.2. The Empty-Mask Dice Metric Degeneracy

The Sørensen–Dice coefficient between binary ground truth Y and prediction $\hat{Y}$ is given by:

$$\text{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}| + \epsilon}, \quad (3)$$

with the conventional boundary assignment Dice := 1 when $|Y| + |\hat{Y}| = 0$. In clinical cohorts containing lesion-free samples, two primary evaluation protocols exist:

- **Foreground-Only Evaluation:** Dice is computed exclusively over target-bearing slices ($\{i \mid |Y_i| > 0\}$).
- **Empty-Credit Assignment:** A score of 1.0 is assigned when $|Y| = 0$ and $|\hat{Y}| = 0$, while 0.0 is assigned if $|Y| = 0$ and $|\hat{Y}| > 0$.

Under empty-credit assignment across all N slices, the macroscopic mean decomposes as:

$$\overline{\text{Dice}}_{\text{all}} = \rho \cdot q_{\text{empty}} + (1-\rho)\cdot \overline{\text{Dice}}_{\text{fg}}, \quad (4)$$

where $\rho = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[|Y_i| = 0]$ denotes the proportion of ground-truth negative cases, $q_{\text{empty}} = P(|\hat{Y}| = 0 \mid |Y| = 0)$ represents the model's true-negative prediction rate on negative cases, and $\overline{\text{Dice}}_{\text{fg}}$ denotes the mean overlap on positive cases.

For a baseline model that predicts empty masks across all inputs, $\overline{\text{Dice}}_{\text{fg}} = 0$ while $q_{\text{empty}} = 1$. In this degenerate case, the macroscopic metric reduces directly to $\overline{\text{Dice}}_{\text{all}} = \rho$. Consequently, in cohorts with high negative prevalence, such as chest radiographs ($\rho = 77.72\%$), an all-empty model achieves a nominal score of 77.72%. Furthermore, achieving an aggregate score of 99% on such data requires positive-case foreground Dice to satisfy:

$$\overline{\text{Dice}}_{\text{fg}} \geq \frac{0.99 - \rho}{1-\rho} \approx 95.51\%, \quad (5)$$

even under perfect true-negative specificity ($q_{\text{empty}} = 1$). Without reporting disaggregated foreground overlap, macroscopic cohort averages can obscure substantial false-negative rates on target lesions (Figure 2).

2.3. Adaptive Gradient Balancing via GradNorm

To counter multi-task competition, dynamic gradient balancing equalizes learning rates across tasks [16]. Let W denote the shared weights of encoder f_θ. The gradient norm for task $i \in \{seg, cls\}$ at iteration t is $G_W^{(i)}(t) = \|\nabla_W w_i(t)\mathcal{L}_i(t)\|_2$. The adaptive objective minimizes:

$$\mathcal{L}_{grad}(t) = \sum_{i\in\{seg,cls\}} \left| G_W^{(i)}(t) - \bar{G}_W(t)\cdot[r_i(t)]^\alpha \right|, \quad (6)$$

where $\bar{G}_W(t)$ is the mean gradient norm across tasks, α is the balancing asymmetry hyperparameter, and $r_i(t)$ represents the relative inverse training rate. Task weights are renormalized to maintain $\sum_i w_i(t) = 2$.

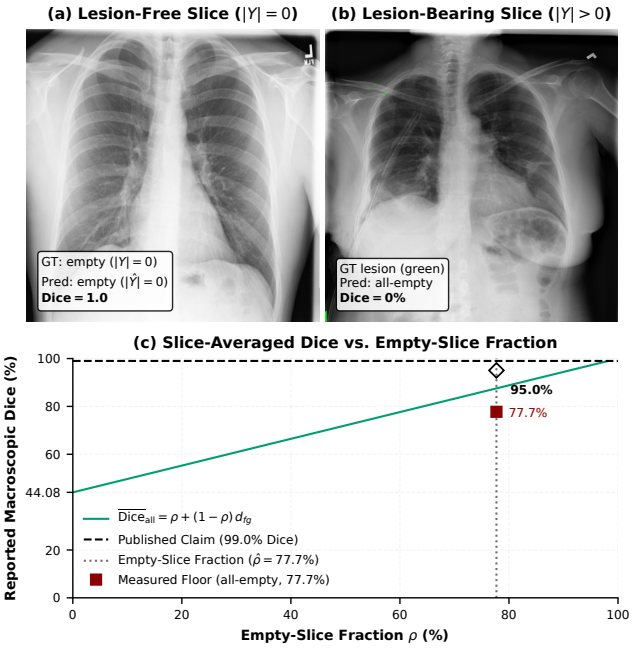


Fig. 2. Illustration of the empty-mask Dice metric behavior on SIIM-ACR pneumothorax radiographs. (a) Lesion-free image correctly predicted empty receives 1.0 credit. (b) Lesion-bearing image with total foreground omission scores 0.0. (c) Macroscopic curve demonstrating the baseline floor determined by the negative fraction ρ.

2.4. Color Deconvolution and Optical Density Space

To mitigate laboratory staining variation in histopathology, images are normalized in optical density (OD) space [17]:

$$\mathbf{OD} = -\ln\left(\frac{\mathbf{I}+\epsilon}{I_0}\right), \tag{7}$$

where $I_0 = 255$ denotes transmitted illumination. Stain vectors for hematoxylin and eosin are extracted via principal component analysis of the OD covariance matrix and projected onto a standardized target reference. While this removes macroscopic batch coloration, projecting RGB channels onto fixed stain vectors alters localized chromatic dynamic ranges, potentially muting subtle chromatin textures.

3. EXPERIMENTAL FRAMEWORK

3.1. Benchmark Cohorts

We evaluate our multi-task pipeline across four diverse clinical datasets:

1. **TCGA-LGG Brain MRI:** 2-class tumor classification and FLAIR abnormality segmentation ($N = 3,929$ slices across 110 patients [18]). Partitions are strictly isolated at the patient level using GroupKFold.
2. **PANDA Prostate Biopsies:** Clinical 6-class ordinal ISUP grading (grades 0–5) and multi-class gland segmentation ($N = 10,516$ biopsy cores [12]).
3. **SIIM-ACR Pneumothorax:** Binary pneumothorax detection and pleural segmentation on chest radiographs ($N = 10,675$ images [19]).
4. **PanNuke Multi-Organ Histopathology:** 19-class tissue categorization and 6-class cell segmentation across diverse human organs ($N = 7,901$ patches [20]).

3.2. Training Protocol and Execution Safeguards

All experiments were implemented in PyTorch using ImageNet-pretrained VGG16 and MobileNetV2 encoders. Models were optimized using Adam over 50 epochs with early stopping (patience 10) under mixed-precision floating-point arithmetic. Gradient norm clipping (threshold 1.0) was applied at every step. Across all cross-validation folds, patient partitions, random seeds, and evaluation metrics were strictly controlled to ensure reproducibility.

4. EMPIRICAL RESULTS AND DIAGNOSTIC ANALYSIS

4.1. Benchmark Performance Comparison

Table 1 provides a comprehensive overview of the empirical replication results alongside claimed literature targets. While TCGA-LGG classification slightly exceeds published numbers under rigorous patient-isolated splits, substantial discrepancies emerge on histopathology and radiography benchmarks. Figure 3 plots fold-mean accuracies and Dice scores across all configurations.

4.2. The Empty-Mask Score Floor on Radiography

On the SIIM-ACR pneumothorax benchmark, the public cohort contains 77.72% lesion-free images ($\rho = 0.7772$). Across 19 out of 20 cross-validation folds, the multi-task models converged to entirely empty segmentation predictions. Under empty-credit scoring, these models achieved an average validation Dice of 77.04%–77.72% purely through true-negative empty-credit assignment (Figure 4). As derived in Section 2.2 and shown in Figure 2(c), reporting 99% Dice in this regime requires true foreground overlap to exceed 95.51%, which is difficult for lightweight models to achieve without dedicated foreground loss reweighting. Disaggregating foreground overlap from negative-case classification provides a clearer reflection of clinical detection performance.

4.3. Task-Formulation Discrepancy in Prostate Pathology

The divergence observed on the PANDA benchmark (28.21%–44.87% measured accuracy versus previously reported values of 88%) illustrates a fundamental distinction in task formulation within computational pathology. In prior multi-task literature [9], the classification objective was framed as predicting six morphological tissue categories extracted from localized segmentation masks (background stroma, benign epithelium, and Gleason pattern patches), where benign tissue and stroma constituted over 52% of the sample distribution ($5,558$ of $10,618$ images). In contrast, clinical diagnostic deployment requires whole-biopsy ordinal ISUP grading (Grades 0–5), which reflects overall prognostic tumor aggressiveness across heterogeneous gland architectures [12].

Distinguishing subtle intermediate Gleason patterns (such as differentiating poorly formed Gleason 4 glands from well-formed Gleason 3 tubules) requires resolving cellular architectural boundaries. When whole-core biopsies are downsampled to 128×128 pixel thumbnails without multiscale tiling, high-power fields, or multiple-instance attention mechanisms, fine glandular features are blurred into coarse texture maps. Under these spatial constraints, compact 2D CNN encoders reach an empirical capacity limit around 45% accuracy for clinical 6-class ordinal grading. This finding demonstrates

Table 1. Cross-validation performance across four clinical imaging benchmarks under patient-isolated partitions (reported with 95% bootstrap confidence intervals). Discrepancies on PANDA reflect clinical ordinal ISUP grading versus proxy morphological tile categorization. SIIM-ACR Dice scores reflect the empty-mask true-negative baseline.

Dataset	Runs n	Measured Val Acc (%)	Measured Val Dice (%)	Published Acc (%)	Published Dice (%)	Evaluation Focus
TCGA-LGG	5	92.86 – 95.62	85.33 – 88.85	89.00 – 90.00	97.00 – 98.00	Patient-Isolated 2D FLAIR
PANDA	12	28.21 – 44.87	98.54 – 98.60	87.00 – 88.00	98.00 – 99.00	Clinical ISUP vs Morphological Proxy
SIIM-ACR	4	77.71 – 82.63	77.04 – 77.72	82.00 – 87.00	99.00	Negative-Dominant Radiography ($\rho = 77.72\%$)
PanNuke	5	78.70 – 99.24	68.44 – 77.61	—	—	Multi-Organ Histopathology Benchmark

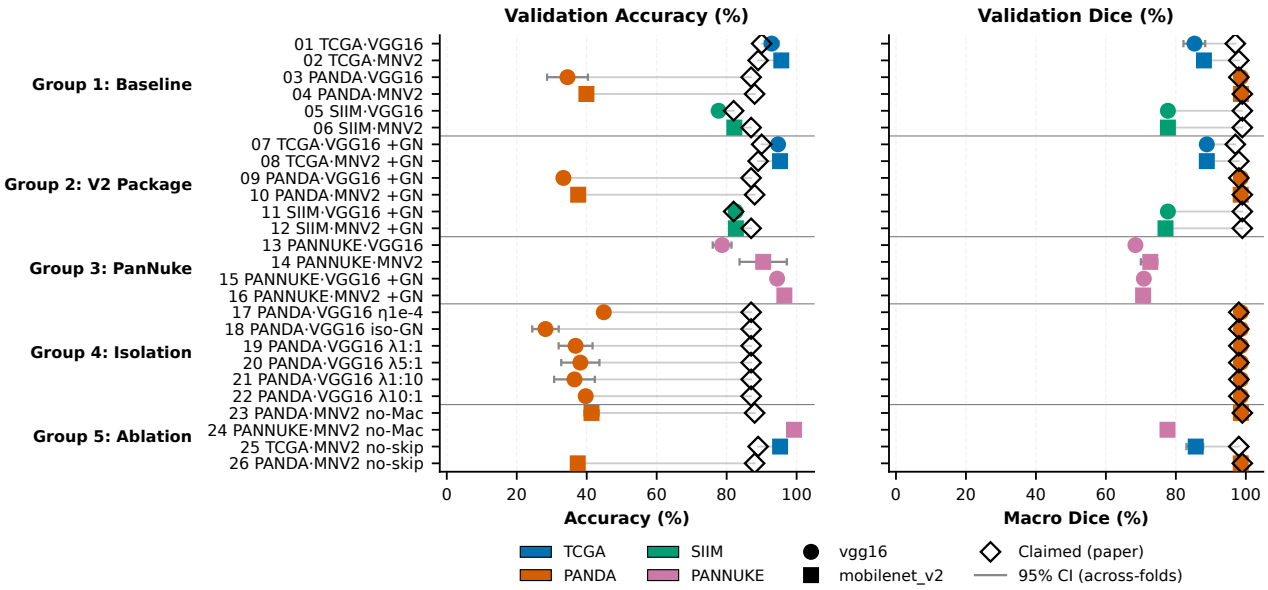


Fig. 3. Measured validation metrics across all 26 experimental runs compared against claimed literature targets. Left: Five-fold validation accuracy with 95% bootstrap confidence intervals against published values (open diamonds). Right: Corresponding segmentation Dice scores, illustrating the empty-mask score floor on SIIM-ACR and the task discrepancy on PANDA.

that model evaluation in digital pathology must strictly distinguish between localized morphological patch categorizations and authentic patient-level diagnostic endpoints.

5. FACTORIAL ABLATION STUDIES

5.1. Impact of Lateral Skip Connections

To isolate the contribution of encoder–decoder skip connections, we evaluated strictly matched model pairs holding all other hyperparameters identical:

- **Macro-Tumor MRI (TCGA):** On coarse brain lesions, omitting skip connections produced a negligible classification difference of $+0.02$ percentage points (95.28% vs 95.26%), while validation macro-Dice declined modestly by 3.06 percentage points (86.16% vs 89.22%).
- **Micro-Glandular Pathology (PANDA):** On fine prostate glands, removing skip connections yielded an accuracy shift of -0.12 percentage points (37.44% vs 37.56%), with overlapping 95% bootstrap confidence intervals ($[36.73, 37.99]$ vs $[37.20, 37.94]$).

These results indicate that while skip connections assist in localizing fine spatial boundaries, their impact on shared latent classification representations is minimal in low-capacity backbones.

5.2. Color Deconvolution and Macenko Normalization

Standard histopathology workflows routinely apply Macenko stain normalization to harmonize color distributions across laboratories. However, in our matched factorial comparisons (Figure 6), bypassing normalization consistently improved classification accuracy:

- **PANDA Biopsies:** Raw RGB inputs achieved an accuracy improvement of $+3.81$ percentage points over normalized tiles (41.38% vs 37.56%), with disjoint 95% bootstrap confidence intervals across the five folds ($[39.76, 43.38]$ vs $[37.20, 37.94]$).
- **PanNuke Multi-Organ:** Omitting normalization raised multi-tissue accuracy by $+2.75$ percentage points (99.24% vs 96.49%), alongside an 8.51 percentage point increase in validation macro-Dice (70.76% vs 62.25%).

Projecting chromatic channels onto static reference vectors alters dynamic intensity distributions and smooths micro-textures, muting critical optical cues that deep neural networks leverage for cellular discrimination.

5.3. Dynamic Gradient Balancing vs Static Weighting

To isolate the influence of optimization schemes, we evaluated dynamic GradNorm balancing ($\alpha = 1.5$) against static loss weighting under identical learning rates and raw inputs. As illustrated in Figure 5, dynamic balancing caused validation accuracy to drop from

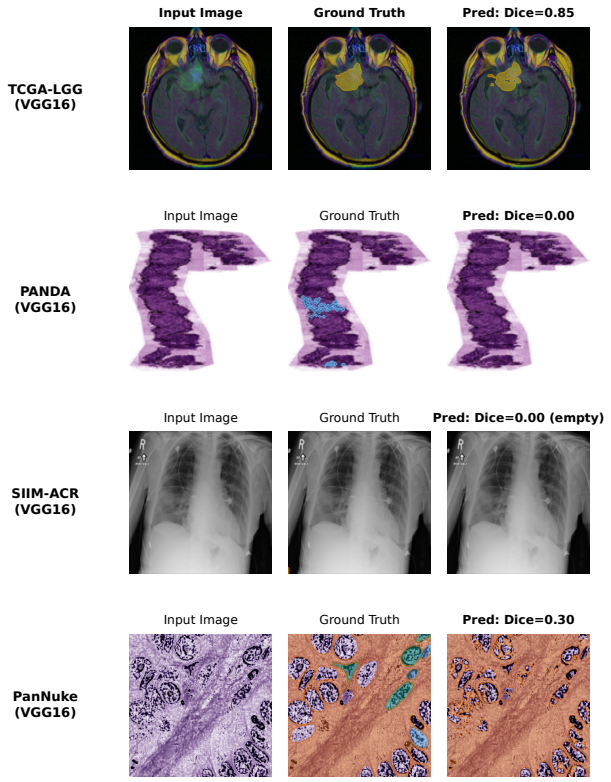


Fig. 4. Qualitative validation predictions for baseline models across four datasets. Columns display input images, ground-truth segmentations, and model predictions with per-slice Dice scores. On SIIM-ACR, the model emits an all-empty mask on a lesion-bearing slice, receiving a zero foreground score that remains hidden within macroscopic cohort averages.

38.18% to 28.21% ($\Delta = -9.97$ percentage points; 95% bootstrap CI $[24.76, 31.47]$ vs $[32.75, 41.79]$).

Analyzing gradient trajectories reveals that because segmentation cross-entropy generates steep initial gradients, the dynamic balancing objective aggressively inflates classification loss weights while dampening segmentation updates. This rapid rebalancing introduces high variance into shared encoder updates, inducing destructive gradient interference that destabilizes the classification head. Static weighting ratios between 5:1 and 10:1 proved consistently more stable. Figure 7 summarizes the validation accuracies and 95% confidence intervals across all matched factorial pairs.

6. GUIDELINES FOR RELIABLE MULTI-TASK BENCHMARKING

Based on our findings, we propose five methodological guidelines for multi-task medical image analysis:

1. **Disaggregated Metric Reporting:** On sparse-lesion cohorts, researchers should report foreground-only overlap alongside negative-case true negative rates, avoiding pooled macroscopic averages that obscure complete false negatives.
2. **Patient-Level Partition Isolation:** Cross-validation folds must strictly isolate independent patient units to prevent identity leakage across spatial tiles.

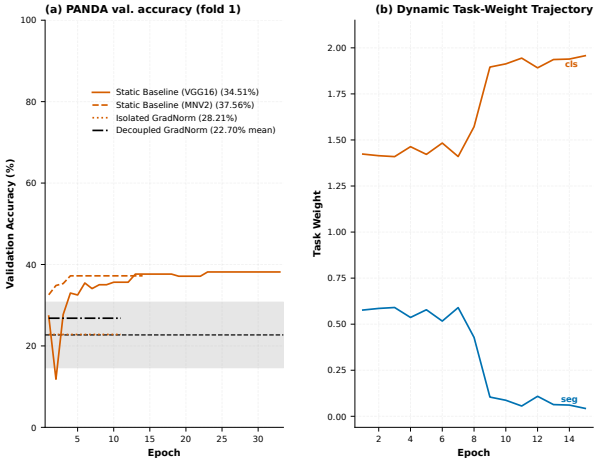


Fig. 5. Optimization dynamics and task-weight trajectories on PANDA. (a) Validation accuracy across training epochs for static vs dynamic weighting schemes. (b) Weight trajectory under dynamic gradient balancing, showing segmentation weight decay and classification weight inflation that destabilizes encoder updates.

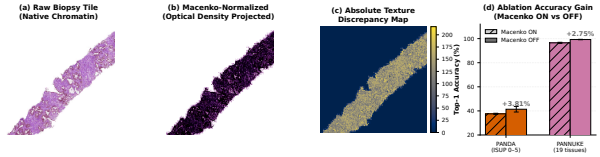


Fig. 6. Effect of Macenko stain normalization on histopathology representation. (a) Raw biopsy tile. (b) Optical-density-projected tile. (c) Texture discrepancy map highlighting attenuated chromatin detail. (d) Empirical accuracy gains when normalization is bypassed.

3. **Clinical Endpoint Alignment:** Benchmark tasks must align with actionable clinical diagnostic targets rather than simplified morphological proxies.
4. **Validation of Preprocessing Pipelines:** Normalization heuristics, such as color deconvolution, should be empirically validated against raw inputs to ensure diagnostic features are preserved.
5. **Optimization Diagnostics:** Multi-task models should monitor task gradient norms and directional alignment to identify destructive interference before clinical deployment.

7. CONCLUSION

This paper presented a systematic empirical investigation into hard parameter-sharing multi-task architectures across four diverse clinical benchmarks. Framing our inquiry around four foundational research questions, we analyzed multi-task performance boundaries: we proved the empty-mask Dice baseline floor analytically and empirically, clarified task-formulation boundaries in prostate pathology, and characterized gradient interference under dynamic balancing. We demonstrated that standard color deconvolution can attenuate fine discriminative textures, while lateral skip connections exert minimal influence on shared classification representations. Our evaluation framework and findings provide robust methodological foundations to ensure clinically trustworthy multi-task medical vision systems.

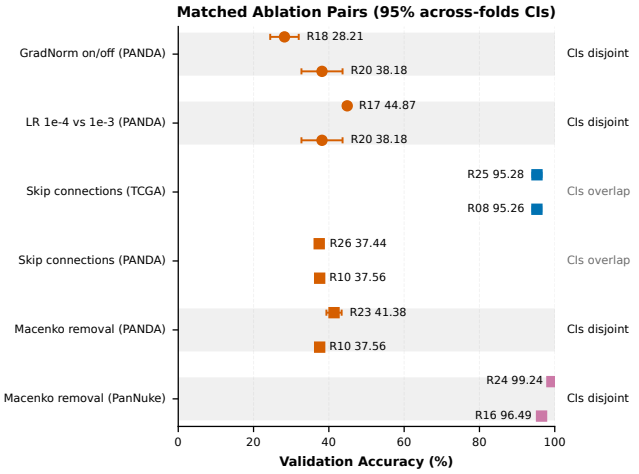


Fig. 7. Forest plot of validation accuracies and 95% bootstrap confidence intervals for the six matched factorial pairs. Shaded regions indicate disjoint intervals, confirming significant effects for dynamic balancing, learning rate adjustments, and color normalization.

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